

AI-Powered Smart Event Management and Analytics System

Project Proposal — Academic Project Document

Project Title	AI-Powered Smart Event Management and Analytics System
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Problem Statement

Event organizers currently manage registrations, ticketing, attendance, and feedback with no data-driven way to predict attendance or ticket demand. This project builds a web-based system where organizers manage events while an ML layer predicts expected attendance and demand from historical data, and users receive personalized event recommendations.

Proposed System

Historical event data — category, price, registrations, attendance, cancellations, revenue, and user interactions — trains ML models to predict attendance and demand, and powers a recommendation engine. Statistical analysis (descriptive stats, correlation, probability, significance testing) surfaces which factors drive attendance. Everything is delivered through a full-stack web app with a REST API, interactive dashboard, and SQL database, built following standard Software Engineering practices.

Objectives

- Build a full event management system: create, update, and manage events.
- Let users discover events, register, book tickets, and manage bookings.
- Build and compare ML models for attendance and ticket-demand prediction.
- Develop a personalized event recommendation feature.
- Evaluate models using accuracy, precision, recall, F1, or MAE as fits the task.
- Compute descriptive statistics and correlations across price, category, and attendance.
- Estimate probabilities of attendance and high-demand events.
- Deliver an interactive dashboard and apply SE practices (UML, testing, documentation).

Scope

In scope: event creation and management, registration and authentication, ticket booking with QR-based check-in, ML-based attendance/demand prediction, recommendations, statistical and probability analysis, an interactive dashboard, REST API, SQL database, and SE documentation (UML, testing).

Out of scope: national-scale deployment, deep learning beyond course level, fraud detection, real-time streaming infrastructure, and a native mobile app.

Technology Stack

- ML: Python, pandas, NumPy, scikit-learn — Logistic Regression, Decision Tree, Random Forest, KNN, Naive Bayes.
- Statistics: NumPy, pandas, SciPy — descriptive stats, correlation, probability, significance testing.
- Visualization: Matplotlib, Plotly, Streamlit, Power BI or Tableau.
- Backend: Flask or Django (REST API). Frontend: React or HTML/CSS/JS.
- Database: SQL (MySQL/PostgreSQL/SQLite). SE Tools: Git/GitHub, UML, Draw.io, Postman.

Expected Outcomes

- A trained model predicting event attendance or ticket demand.
- A personalized recommendation feature and probability-based demand insights.
- A complete full-stack system for discovery, registration, and ticket booking.
- An interactive dashboard covering registrations, sales, revenue, and predictions.
- QR-based ticketing and check-in, with a well-documented, tested, modular system.

Timeline

Phase	Focus	Activities
Week 1	Setup	Design DB schema, collect/prepare event data, run EDA.
Week 2	Statistics	Descriptive stats, correlation and probability analysis.
Week 3	Machine Learning	Build and compare attendance/demand and recommendation models.
Week 4	Full Stack	Build frontend, backend REST API, database, booking modules.
Week 5	Dashboard	Integrate ML predictions; build the analytics dashboard.
Week 6	Testing & Docs	Unit/integration testing, UML documentation, demo prep.